University of Colombo, Sri Lanka

UCSC *University of Colombo School of Computing*

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination — Semester II– UCSC AY21 [held in March/April 2025]

SCS1310 — Object Oriented Modelling and Programming

(Two (2) Hours)

Answer ALL questions

Number of Pages = 16 Number of Questions = 4

To be completed by the candidate								
Index Number								

Important Instructions to candidates:

- Students should answer in the medium of English language only using the space provided in this question paper.
- Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
- Write your index number CLEARLY on each and every page of this question paper.
- The duration of the paper is Two (2) hours.
- This paper has 4 questions in 16 pages (including the Cover Page).
- Answer all the questions.
- Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.
- Do not tear off any part of this question paper. Under no circumstances may this paper, used or unused, be removed from the Examination Hall by a candidate.

To be completed by the examiners

1	
2	
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1. Consider the following Book Class Written in C++.

```cpp
#include <iostream>
using namespace std;

class Book {
private:
    string *title;
    string author;
    float price;

public:
    Book(); // -------------------------- (I)
    Book(string *, string, float); // ---- (II)
    Book(const Book &b); // --------------(III)
    ~Book(); // ------------------------- (IV)

    void display(); // ------------------- (V)

    Book& operator=(const Book &b) { // ----(VI)
        if (this != &b) {
            delete title;
            title = new string(*b.title);
            author = b.author;
            price = b.price;
        }
        cout<<"Inside operator function\n\n";
        return *this;
    }
}book1;
```

(a). Identify the most appropriate name for each of the marked signatures from (I) to (VI) using the following list of words:

`Member Function, Non-Member Function, Copy Constructor, Operator Overloading Function, Default Constructor, Friend Function, Shallow Copy Constructor, Parameterized Constructor, Function Overloading, Destructor, Delete Function, Malloc Function`

[3 marks]

(I).	(IV).
(II).	(V).
(III).	(VI).

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(b). Briefly explain the functionality of the above identified signatures from (I) to (VI). The definition of each signature is provided in Figure below.

```
Book::Book() {
    title = new string("Unknown");
    author = "Unknown";
    price = 0.0;
    cout << "1st Constructor called\n";
}

Book::Book(string *t, string a, float p) {
    title = new string(*t);
    author = a;
    price = p;
    cout << "2nd Constructor called\n";
}

Book::Book(const Book &b) {
    title = new string(*b.title);
    author = b.author;
    price = b.price;
    cout << "3rd Constructor called\n";
}

Book::~Book() {
    cout << "~Book is called for: " << *title << "\n";
    delete title;
}

void Book::display() {
    cout << "Book Details:\n";
    cout << "Title: " << *title << ", Author: " << author
    << ", Price: " << price << "\n\n";
}
```

[6 marks]

(I).

(II).

(III).

(IV).

(V).

(VI).

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(c). What will be the output of the given code if the `main()` function is provided as shown in the Figure below?

```
int main() {
    Book book2;

    string name = "Secret Seven";
    Book book3(&name, "Enid Blyton", 395.00);

    Book book4 = book3;

    book2 = book3;

    book1.display();
    book2.display();
    book3.display();
    book4.display();

    return 0;
}
```

[6 marks]

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(d). Briefly explain the purpose of explicitly using `delete title;` in the item marked as (IV). What would happen if this statement is omitted?

[5 marks]

(e). Suppose the signature denoted by (III) and its definition are removed from the class. What issue would arise when executing `Book book4 = book3;`?
Explain the problem with reference to memory allocation.

[5 marks]

(f). Suppose the function marked as (VI) is replaced with a friend function and defined outside the class. What would be the output of the code upon execution?
Justify your answer.

[5 marks]

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2. **Only ONE** answer is correct in the following 10 MCQs listed from (a) to (j). Consider the case study below and underline the most appropriate answer.

[2 x 10 = 20 marks]

Case Study: Rent-a-Car System

The Rent-a-Car System enables customers to browse available vehicles, make reservations, and return rented cars. Customers can search for cars based on model, price, and availability. Once they find a suitable car, they can proceed with a reservation by providing personal details and payment information. The system verifies the booking and confirms the reservation. Upon arrival for pickup, the rental agent verifies the customer's documents before handing over the car. For VIP customers, the system offers priority service, allowing them to bypass standard verification procedures, receive exclusive discounts, and access premium vehicles. VIP customers also have the option of personalized assistance, such as dedicated agents and express checkout services. At the end of the rental period, the customer returns the car. The rental agent inspects the vehicle for damages before closing the rental agreement. If any damage is found, additional charges may be applied. The system also allows administrators to manage vehicle availability, pricing, and customer records, ensuring smooth operation and customer satisfaction.

(a). Which of the following relationship best describes the connection between "Process Payment" usecase and "Customer"?

i. Generalization
ii. Association
iii. Include
iv. Extend
v. Dependency

(b). If a new feature is introduced where customers can cancel a reservation before the rental period starts, which of the following best represents the relationship between "Make a Reservation" and "Cancel Reservation"?

i. Generalization – "Cancel Reservation" is a specialized form of "Make a Reservation"
ii. <<extend>>– "Cancel Reservation" is an optional feature dependent on "Make a Reservation"
iii. <<include>>– "Make a Reservation" must always include "Cancel Reservation"
iv. Aggregation – "Cancel Reservation" and "Make a Reservation" share the same attributes
v. No relationship – These use cases are independent and do not interact

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(c). In a sequence diagram, what would be the most appropriate way to represent an optional action, such as applying a discount if the customer is a VIP?

i. Use an alt (alternative) fragment to show the conditional execution.
ii. Represent it as a separate object that handles the discount process.
iii. Use an include relationship between the "VIP customer" and the "apply discount" usecase.
iv. Draw a loop fragment to repeat discount calculation until VIP status is confirmed.
v. Represent the action with a return message from the discount service.

(d). In a sequence diagram for returning a rented car, which of the following statements about the destroy message (X marker) is correct?

i. It indicates that the object is permanently deleted from the system.
ii. It ends the interaction without removing the object from memory.
iii. It represents a cancellation action where the object is paused temporarily.
iv. It shows the object completes its role but may be reactivated later.
v. It is used to indicate an error in the transaction process.

(e). In the state transition diagram for the Rent-a-Car System, which of the following transitions is most appropriate when a customer does not pick up the reserved car within the allowed time?

i. "Reserved" → "Available"
ii. "Reserved" → "Rented"
iii. "Reserved" → "Under Maintenance"
iv. "Rented" → "Available"
v. "Available" → "Lost"

(f). Consider the following `Assertion (A)` and the `Reason (R)` pair and identify the correct answer.

Assertion (A): A rental car in the Under Maintenance state can transition directly to Rented if a VIP customer requests the same car.
Reason (R): VIP customers are allowed to override standard rental restrictions.

i. Both (A) and (R) are true, and (R) is the correct explanation of (A).
ii. Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).
iii. (A) is true, but (R) is false.
iv. (A) is false, but (R) is true.
v. Both (A) and (R) are false.

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(g). Consider the following activity sequence.

A. Customer → Search Cars → View Car Details → Make Payment
B. Customer → Return Car → Inspection → Damage Found → Pay Fine
C. Admin → Update Pricing → Save Changes → Logout
D. Customer → Request VIP Service → Eligibility Check → Update VIP Status

In an Activity Diagram for the Rent-a-Car System, which of the above sequence(s) is/are most likely to model using a diamond shaped symbol?

i. (B) Only.
ii. (A) and (C) Only.
iii. (B) and (D) Only.
iv. (B), (C) and (D) Only.
v. None of the above.

(h). Which of the following scenarios in the Rent-a-Car System demonstrates parallel execution using a fork in an Activity Diagram?

i. A customer searches for a car, selects a model, and then proceeds to payment.
ii. The system updates car availability while sending a confirmation email to the customer.
iii. The rental agent either verifies documents or marks the reservation as pending based on missing details.
iv. The customer chooses either an online or offline payment method to complete the transaction.
v. The system finalizes the reservation only after the rental agreement is signed.

(i). Consider the following three statements.

(A). Cars and Reservations are unrelated.
(B). A Reservation is a type of Car (inheritance).
(C). A Customer can have multiple Reservations, but each Reservation is linked to only one Car.

Which of the above statements is/are correct regarding the classes Car, Reservation, and Customer in a Rent a Car Class Diagram?

i. (A) Only.
ii. (B) Only.
iii. (C) Only.
iv. (A), and (B) Only.
v. None of the above.

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(j). In the Rent-a-Car System, how should RentalAgreement, Customer, and Car be related in a Class Diagram?

i. Simple associations, with no need for Aggregation or Composition.
ii. Both Customer and Car should have Aggregation with RentalAgreement.
iii. Both Customer and Car should be linked to RentalAgreement using Composition.
iv. Customer and RentalAgreement should have Aggregation, while Car and RentalAgreement should have Composition.
v. Car and RentalAgreement should have Aggregation, while Customer and RentalAgreement should have Composition.

3. (a). Only ONE answer is correct in the following 15 MCQs.
Cross or color the **best suite answer** among the given options in the table below.

[2 x 15 = 30 marks]

i.	(A)	(B)	(C)	(D)	(E)
ii.	(A)	(B)	(C)	(D)	(E)
iii.	(A)	(B)	(C)	(D)	(E)
iv.	(A)	(B)	(C)	(D)	(E)
v.	(A)	(B)	(C)	(D)	(E)
vi.	(A)	(B)	(C)	(D)	(E)
vii.	(A)	(B)	(C)	(D)	(E)
viii.	(A)	(B)	(C)	(D)	(E)

ix.	(A)	(B)	(C)	(D)	(E)
x.	(A)	(B)	(C)	(D)	(E)
xi.	(A)	(B)	(C)	(D)	(E)
xii.	(A)	(B)	(C)	(D)	(E)
xiii.	(A)	(B)	(C)	(D)	(E)
xiv.	(A)	(B)	(C)	(D)	(E)
xv.	(A)	(B)	(C)	(D)	(E)

i. Consider the following three statements regarding the `Inheritance` in OOP.

I. It is a mechanism for creating new classes from existing ones.
II. It is a mechanism that allows code reuse.
III. Train and Carriages has the relationship define as Inheritance.

Which of the above statement(s) is/are TRUE?

A. I only. B. II only. C. I and II only.
D. II and III only. E. I, II, and III.

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ii. Which of the following is TRUE according to the following class definition?

```
class A : private B, public C {
// body of the class
}
```

A. A cannot access any data in B. B. only public data in C will become public to A. C. All public, protected and private data in B will become private to A. D. Both public and protected data in C will become private to B. E. Only the private and protected data in B will become private to A.

iii. Consider the following 3 statements regarding `inheriting constructors` in C++.

I. Base class constructors can call derived class constructors.
II. Derived class will not inherit the constructors from the base class by default.
III. If more than one constructor available on the base class, all of them will be called sequentially.

Which of the above statement(s) is/are TRUE?

A. I only.	B. I and II only.	C. II only.
D. II and III only.	E. I, II, and III.	

iv. Consider the following 3 statements regarding the `Diamond Problem`.

I. It can be occurred when a class have two child classes.
II. You always get a runtime error when it occurred.
III. `virtual` keyword can be used to prevent the diamond problem.

Which of the above statement(s) is/are TRUE?

A. I only.	B. I and II only.	C. III only.
D. I and III only.	E. I, II, and III.	

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v. What is the output of the following program written in C++?

```
class A {
  int x = 1;
public:
  void setX(int i) {x = i;}
  void print() {cout << x;} };

class B: virtual public A {
public: B() {setX(3);} };

class C: virtual public A {
public: C() {setX(5);} };

class D: public C, public B {};

int main() {
    D d;
    d.print(); }
```

A. 1	B. 3	C. 5
D. Runtime error	E. Compilation Error	

vi. What is the output of the following program written in C++?

```
class Shape {
  public:
    void area() {cout<<"A";}
    void area(int x) {cout<<"B";}
    void area(char x) {cout<<"C";} };

int main() {
  Shape s;
  s.area(); }
```

A. A	B. B	C. C
D. Compilation Error	E. Run time Error	

vii. Consider the following three statements regarding the `Function Overloading` in OOP.

I. It is the way to achieve the compile time polymorphism.
II. It is not mandatory to use the same name to overload a function.
III. You can not change the parameter list when you overload a function.

Which of the above statement(s) is/are TRUE?

A. I only.	B. II only.	C. I and II only.
D. II and III only.	E. I, II, and III.	

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viii. Following is the definition of a function in a class named ABC written in C++.

```
virtual void func(int x=0, int y=0) = 0;
```

Which of the following is FALSE regarding the above code?

A. This function returns 0 when it calls.
B. You cannot create objects from ABC.
C. You will get a compiler error if you remove the keyword virtual.
D. It is not mandatory to pass 2 integers when calling this function.
E. ABC is an Abstract class.

ix. The purpose of the `override` keyword in C++ is to,

A. define a new abstract method.
B. specify the access level of a method.
C. prevent a method from being overridden.
D. indicate the virtual method which should be overloading.
E. indicate that a derived class method is intended to override.

x. What is the primary purpose of `Abstraction` in OOP?

A. To enhance code readability.
B. To ensure data integrity and security.
C. To enable polymorphism and inheritance.
D. To facilitate code reuse at the implementation.
E. To hide complex implementation details and show only the essential features to the user.

xi. Consider the following three statements regarding the `templates` in C++.

I. It is a feature of C++ that allows to write one code for different data types.
II. It supports generic programming.
III. It is an example of compile time polymorphism.

Which of the above statement(s) is/are TRUE?

A. I only.	B. II only.	C. II and III only.
D. I and III only.	E. I, II, and III.	

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xii. Consider the following definition of a template.

```
template <typename T, char y>
T f(T x) { return x * y; }
```

Which of the following is NOT a correct call of the function `f()`?

A. f<int, 'A'>(10)	B. f<char, 10>('A')	C. f<int, 10>(3.3)
D. f<int, 3.3>(10)	E. f<int, 'A'>(3.3)	

xiii. What is the output of the following program written in C++?

```
int main() {
    try {throw 'a'; }
    catch(...){cout << "X";}
    catch(int x){cout << "Y";}
    catch(char y){cout << "Z";}
}
```

A. X	B. Y	C. Z
D. Compilation Error	E. Run time Error	

xiv. Consider the following three statements regarding the `exception handling` in C++.

I. It is used to manage logical errors of the code.
II. It helps to create more robust and error-resistant programs.
III. You need to write all of your code inside the `try` block in order to trigger an exception.

Which of the above statement(s) is/are TRUE?

A. I only.	B. I and II only.	C. II only.
D. II and III only.	E. I, II, and III.	

xv. Consider the following three statements regarding the `Composition` in OOP.

I. It is the same as inheritance in OOP.
II. It represents a `has-a` relationship between two classes.
III. House and Mother has the relationship define as Composition.

Which of the above statement(s) is/are TRUE?

A. I only.	B. I and II only.	C. II only.
D. II and III only.	E. I, II, and III.	

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4. (a). i. What is the output of the following code written in C++?

```
1  #include <iostream>
2  using namespace std;
3
4  class A {
5     public:
6        A() {cout<<"1";}
7        A(int x) {cout<<"x2";}
8        void print() {cout<<"x3";}};
9
10 class B: public A {
11    public:
12       B() {cout<<"x4";}
13       B(int x) {cout<<x;}};
14
15 int main() {
16    B b;
17    b.print(); }
```

[3 marks]

....

ii. What will be the output of the above program when the `line 16` changed as follows?

`B b(5);`

[3 marks]

....

(b). i. What is the output of the following code written in C++?

```
#include <iostream>
using namespace std;

class A {
   public:
      int f(int x) {cout<<x;}
      int f(int x, int y) {cout<<x+y;} };

class B: public A {
   public:
      int f(int x) {cout<<x*x;} };

int main() {
   A a; B b;
   a.f(10,10);
   b.f(10); }
```

[3 marks]

....

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ii. What wil be the outcome of the above program when the `line 15` changed as follows? Explain your answer.

```
b.f(10, 10);
```

[3 marks]

(c). i. What is the output of the following code written in C++?

```
1   #include <iostream>
2   using namespace std;
3
4   class A {
5      public:
6        void print(){cout<<"A";} };
7
8   class B: public A {
9      public:
10       void print(){cout<<"B";} };
11
12  int main() {
13     B b;
14     try { throw b;}
15       catch(B b) {b.print();}
16       catch(A a) {a.print();}
17       catch(...) {cout<<"C";} }
```

[2 marks]

ii. What will be the output of the above program when we swap the `line 15` and the `line 16`?

[2 marks]

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(d). What is the output of the following code written in C++?

```
template <typename T, typename U, int n=1>
U fun(T x, U y) { return (x + y) * n; }

int main() {
  cout << fun<int, char>(2, 'A') << ", ";
  cout << fun<double, int, 10>(2.5, 1) << endl;}
```

[4 marks]

....

****